Architecture vs Resolution in Monocular Depth Estimation: A Speed–Accuracy Trade-off Study

Abstract

Monocular depth estimation provides a low-cost alternative to hardware-based sensing for assistive navigation, but real-time performance on edge devices remains challenging.
This study investigates whether model architecture, rather than input resolution, governs the trade-off between inference speed and structural consistency. A transformer-based model (Depth Anything V2 Small) and a CNN-based model (MiDaS Small) were evaluated under a controlled RTX 2050 protocol.
The transformer maintained low structural error (ROI RMSE < 1.0), while the CNN achieved higher throughput (19.42 FPS) but exhibited severe degradation (ROI RMSE > 100).
The results show that architectural design dominates this trade-off, with no configuration achieving both real-time performance (>15 FPS) and acceptable accuracy, highlighting a key limitation for safety-critical deployment.

Introduction

Monocular depth estimation enables 3D scene understanding using a single RGB image, making it suitable for assistive navigation systems.
While previous work has explored input resolution as a factor influencing performance, the role of model architecture under deployment constraints remains underexplored.
This study shifts focus from input scaling to architectural comparison, evaluating whether transformer-based and CNN-based models exhibit fundamentally different behaviour under identical conditions.
In safety-critical contexts, inference latency must remain below ~200 ms (~15 FPS). Therefore, both speed and structural consistency must be considered simultaneously.

Figure 1. Example RGB input frame **(right)** and corresponding monocular depth prediction **(left)**. Brighter regions indicate closer surfaces within the safety-critical ground Region of Interest (ROI).

Problem Context
Existing work assumes that reducing input resolution improves performance [1], [2]. However, it remains unclear whether such improvements are sufficient when architectural constraints dominate computational cost [3].

Research Gap
To determine whether model architecture, rather than input resolution, governs the trade-off between inference speed and structural consistency under controlled deployment conditions.

Research Objective
This study investigates whether systematic input resolution scaling improves real-time feasibility while preserving depth consistency within regions critical for navigation.

Key Contribution
- Controlled architecture comparison under identical conditions
- Strictly locked reproducible protocol (SHA256, reference depth stack)
- ROI-based safety-critical evaluation
- Cross-device validation (RTX, M1, Raspberry Pi)
- Identification of no Pareto-optimal solution

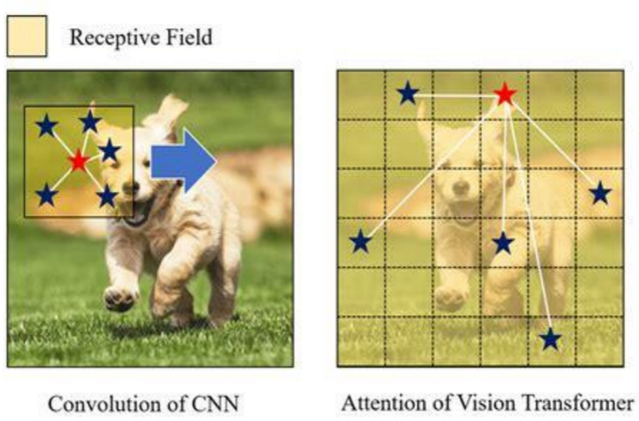


Figure 2. Conceptual comparison of local receptive fields in convolutional networks and global self-attention in Vision Transformers.

Literature Review

CNNs vs. Transformers:
CNN-based approaches are computationally efficient but limited in capturing long-range spatial dependencies, often resulting in weaker global consistency [3], [5].
In contrast, Vision Transformer architectures improve global scene understanding through self-attention mechanisms [4], but introduce significant computational overhead, limiting real-time deployment on edge-constrained systems [5].

Optimization Focus:
Existing optimisation research primarily focuses on model compression, pruning, and knowledge distillation [1], [5]. While input resolution has been widely studied, the comparative impact of model architecture under deployment constraints remains insufficiently explored.

Latency and Safety:
Human auditory and gait termination constraints dictate that system latency exceeding ~200 ms (>15 FPS) drastically increases collision risk for visually impaired users [1], [6], [7]. However, there is limited controlled evaluation of how architectural design influences the trade-off between inference speed and structural consistency under identical deployment conditions.

Test Configuration and Method

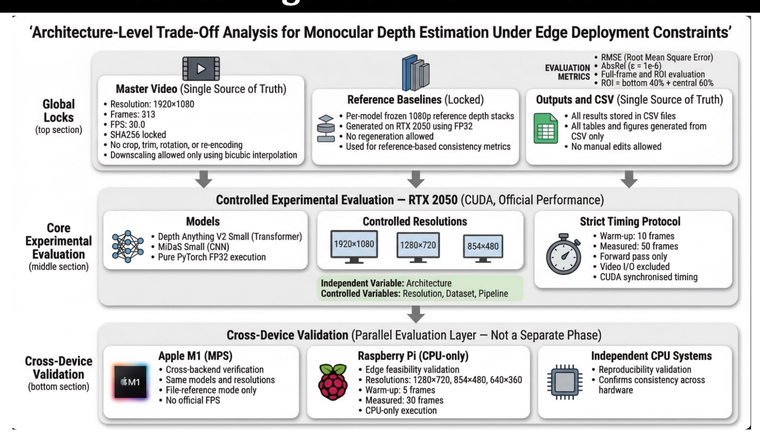
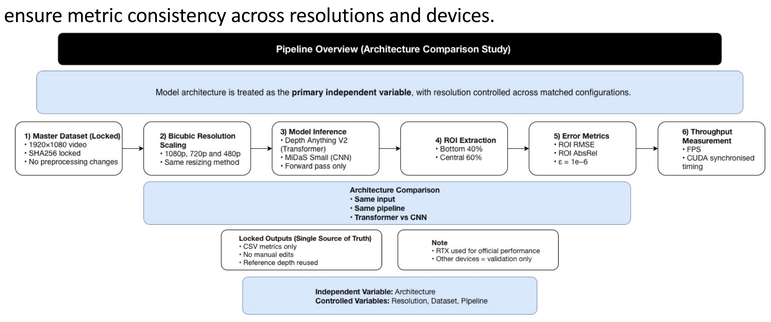


Figure 3. High Level Diagram

All experiments were conducted under a strictly controlled and reproducible protocol in which model architecture was treated as the primary independent variable, while input resolution was held constant across matched configurations.
A single master video (S2_Main_Approach_h264.mp4) was SHA256-locked to prevent dataset drift. A 1920×1080 reference depth stack was generated on an RTX 2050 (CUDA, FP32) and reused in file-reference mode for all non-reference runs to ensure metric consistency across resolutions and devices.



The Region of Interest (ROI) was defined as the bottom 40% and central 60% of each frame to represent safety-critical ground regions. Accuracy was evaluated using ROI RMSE and ROI AbsRel (ε = 1e−6). Bicubic interpolation was applied for all resizing operations, and CSV outputs were treated as the single source of truth.

Experiments were executed on an **NVIDIA RTX 2050 (CUDA backend)** at:
- 1920×1080 (reference baseline)
- 1280×720
- 854×480

Performance was measured over 50 frames using forward-pass-only execution with CUDA-synchronised timing. RTX results form the official benchmark for throughput and trade-off analysis.

Apple M1 (MPS)	Raspberry Pi (CPU-only)	Independent CPU Systems
Cross-backend verification	Edge feasibility validation	Reproducibility validation
Same models and resolutions	Resolutions: 1280×720, 854×480, 640×360	Confirms consistency across hardware
File-reference mode only	Warm-up: 5 frames	
No official FPS reporting	Measured: 30 frames	
	CPU-only execution	

Validation results are reported for consistency and feasibility only and do not redefine the RTX benchmark.

Results

Performance was evaluated using ROI RMSE, ROI AbsRel, and CUDA-synchronised FPS under a controlled architecture comparison protocol.

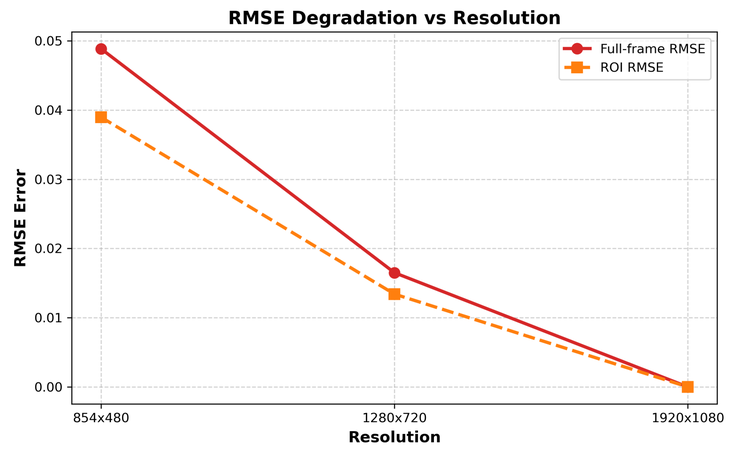


Figure 4. RMSE increases significantly with downsampling, indicating structural degradation in safety-critical regions

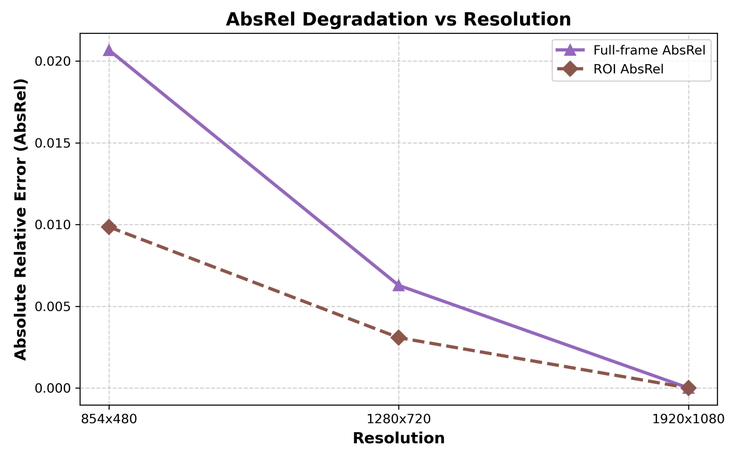


Figure 5. AbsRel error increases with reduced resolution, confirming loss of depth consistency.

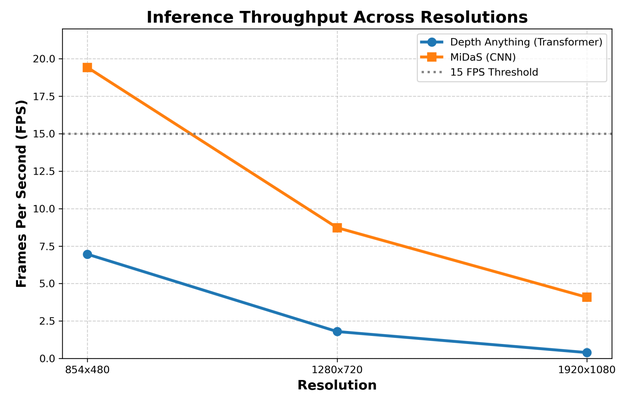


Figure 6. Inference speed increases as resolution decreases. The CNN model reaches a peak of 19.42 FPS at 480p, while the transformer struggles to achieve real-time performance at any scale.

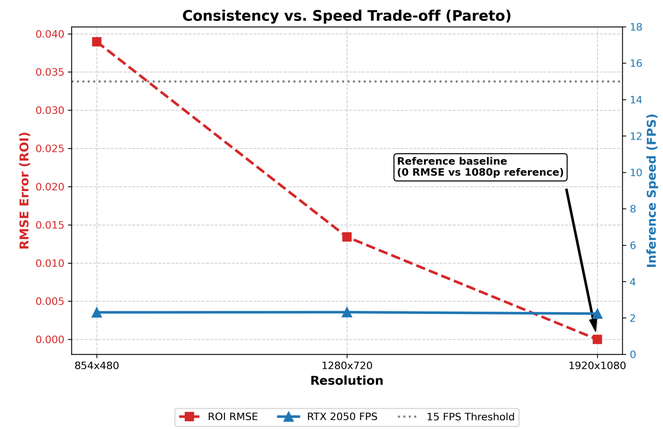


Figure 7. No configuration achieves both low error and high speed, indicating absence of a Pareto-optimal solution.

Key Findings
- ROI error increases as resolution decreases.
- Degradation is monotonic but non-linear.
- ~80% pixel reduction shows **no measurable speed gain**.
- Inference speed remains ~5.4 FPS across all resolutions.
- No configuration reaches the ≥15 FPS real-time threshold.
- No Pareto improvement is observed.

Model	Hardware	Resolution	FPS	RMSE	AbsRel	≥15 FPS	Pareto
RTX 2050 — CUDA benchmark							
Depth Anything	RTX 2050	1920×1080	0.40	0.000	0.000	—	Baseline
Depth Anything	RTX 2050	1280×720	1.80	0.27	0.057	✗	No
Depth Anything	RTX 2050	854×480	6.96	0.96	0.191	✗	No
MiDaS	RTX 2050	1920×1080	4.09	0.000	0.000	—	Baseline
MiDaS	RTX 2050	1280×720	8.73	96.84	0.381	✗	No
MiDaS	RTX 2050	854×480	19.42	101.02	0.355	✓	No
Raspberry Pi 4 — CPU feasibility							
Depth Anything	RPi 4	1280×720	0.024	0.35	0.060	✗	No
Depth Anything	RPi 4	640×360	0.165	0.58	0.116	✗	No
MiDaS	RPi 4	1280×720	0.052	107.97	0.396	✗	No
MiDaS	RPi 4	640×360	0.212	135.92	0.548	✗	No

Table 1. Combined results summary (RTX 2050 & Raspberry Pi 4)

Edge Feasibility
- Raspberry Pi (CPU-only) < 0.25 FPS
- Far below ≥15 FPS safety threshold
- Severe degradation at lower resolutions
- CPU-only deployment is **not viable**

Conclusion and Future Work

This study presents a controlled evaluation of monocular depth estimation under a strictly locked and reproducible protocol, demonstrating that input resolution scaling alone does not improve real-time feasibility. While downsampling reduces spatial detail, it introduces progressive degradation in ROI depth consistency without measurable gains in inference speed on GPU hardware.
The results show that performance is constrained by architectural overhead rather than pixel count. Across all configurations, inference speed remained effectively constant (~5.4 FPS), and no configuration satisfied the real-time safety threshold (≥15 FPS). CPU-only deployment was found to be infeasible under the evaluated conditions.
These findings indicate that the speed–consistency trade-off is fundamentally limited by model architecture. Resolution scaling alone cannot deliver deployable performance for safety-critical assistive navigation.
Future work will focus on architectural optimisation, including lightweight transformer variants and hybrid approaches, evaluated under the same controlled protocol to determine whether meaningful improvements in the speed–consistency trade-off can be achieved.

References

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